Lecture #8: Joint Distributions

MATH 321 – Mathematical Statistics

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Moment-Generating Functions

For any random variable X, the **moment-generating function (MGF)** $M_X(t)$ of a random variable X is:

$$M_X(t) = E[e^{tX}].$$

Moment-Generating Function Example

Example:

1. Find the moment-generating function for $X \sim \text{Exponential}(\lambda)$.

Moment-Generating Function Example

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1. Find the moment-generating function for $X \sim \text{Exponential}(\lambda)$.
2. Find the moment-generating function for $X \sim \text{Bernoulli}(p)$.

Moment-Generating Function Example

Example:

1. Find the moment-generating function for $X \sim \text{Exponential}(\lambda)$.
2. Find the moment-generating function for $X \sim \text{Bernoulli}(p)$.
3. Find the moment-generating function for $X \sim \text{Binomial}(n, p)$.

Properties of Moment-Generating Functions

Theorem: The moment-generating function has the properties that:

1. $M_X(0) = 1$.
2. $\left.\dfrac{d}{ds} M_X(s)\right|_{s=0} = E[X]$.
3. $\left.\dfrac{d^2}{ds^2} M_X(s)\right|_{s=0} = E[X^2]$.
4. $\left.\dfrac{d^k}{ds^k} M_X(s)\right|_{s=0} = E[X^k]$, for any positive integer k.

Example: Given that the MGF of $X \sim \text{Exponential}(\lambda)$ is $M_X(t) = (1 - \beta t)^{-1}$, prove that $E[X] = \beta$ and $V(X) = \beta^2$ using it.

Properties of Moment-Generating Functions

Theorem: Let X and Y be independent random variables. Let $Z = X + Y$. Then

$$M_Z(t) = M_X(t)M_Y(t).$$

Corollary: Consider independent random variables $X_1, \ldots, X_N$. Let $Z = \sum_{i=1}^{N} X_i$ be the sum of random variables. Then the MGF of Z is

$$M_Z(t) = \prod_{i=1}^{N} M_{X_i}(t).$$

What's the Deal with the MGF?

The moment-generating function of a random variable uniquely defines the distribution.

Theorem: Let $M_X(t)$ and $M_Y(t)$ denote the moment-generating functions of random variables X and Y. If both moment-generating functions and $M_X(t) = M_Y(t)$ for all $t \in \mathbb{R}$, then X and Y have the same probability distribution.

The moment-generating function does not exist for all distributions. For example, the **Cauchy distribution** does not have an MGF.

Moment-Generating Functions of i.i.d. Random Variables

Theorem: Let $M_X(t)$ and $M_Y(t)$ denote the moment-generating functions of random variables X and Y. If both moment-generating functions and $M_X(t) = M_Y(t)$ for all $t \in \mathbb{R}$, then X and Y have the same probability distribution.

If these independent random variables are further assumed to be identically distributed, the moment-generating function is

$$M_Z(t) = \left(M_{X_1}(t)\right)^N.$$

Examples of MGFs of i.i.d. Random Variables

Examples:

1. Let $X_1, \dots, X_N$ be a sequence of i.i.d. Bernoulli random variables with parameter p. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?

Examples of MGFs of i.i.d. Random Variables

Examples:

1. Let $X_1, \dots, X_N$ be a sequence of i.i.d. Bernoulli random variables with parameter p. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?
2. Let $X_1, \dots, X_N$ be a sequence of i.i.d. binomial random variables with parameters n and p. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?

Examples of MGFs of i.i.d. Random Variables

Examples:

1. Let $X_1, \dots, X_N$ be a sequence of i.i.d. Bernoulli random variables with parameter p. Let $Z = X_1 + \dots + X_N$. What is the distribution of Z?
2. Let $X_1, \dots, X_N$ be a sequence of i.i.d. binomial random variables with parameters n and p. Let $Z = X_1 + \dots + X_N$. What is the distribution of Z?
3. Let $X_1, \dots, X_N$ be a sequence of independent Poisson random variables with parameter λ_i. Let $Z = X_1 + \dots + X_N$. What is the distribution of Z?

Examples of MGFs of i.i.d. Random Variables

Examples:

1. Let $X_1, \ldots, X_N$ be a sequence of i.i.d. Bernoulli random variables with parameter p. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?
2. Let $X_1, \ldots, X_N$ be a sequence of i.i.d. binomial random variables with parameters n and p. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?
3. Let $X_1, \ldots, X_N$ be a sequence of independent Poisson random variables with parameter λ_i. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?
4. Let $X_1, \ldots, X_N$ be a sequence of independent normal random variables with parameters μ_i and σ_i^2.. Let $Z = X_1 + \cdots + X_N$. What is the distribution of Z?

The Characteristic Function

There is a related function that does exist for all distributions, the characteristic function $\varphi_X(t)$.

For any random variable X, the **characteristic function** $\varphi_X(t)$ of a random variable X is:

$$\varphi_X(t) = E[e^{itX}].$$

Properties of the Characteristic Function

Properties:

1. $\varphi_X(-it) = M_X(t)$.
2. $\displaystyle \frac{d^k}{ds^k} i^k \varphi_X(s)\bigg|_{s=0} = E[X^k]$, for any positive integer k.
3. Consider independent random variables $X_1, \ldots, X_N$. Let $Z = \sum_{i=1}^N X_i$ be the sum of random variables. Then the MGF of Z is

$$\varphi_Z(t) = \prod_{i=1}^N \varphi_{X_i}(t).$$